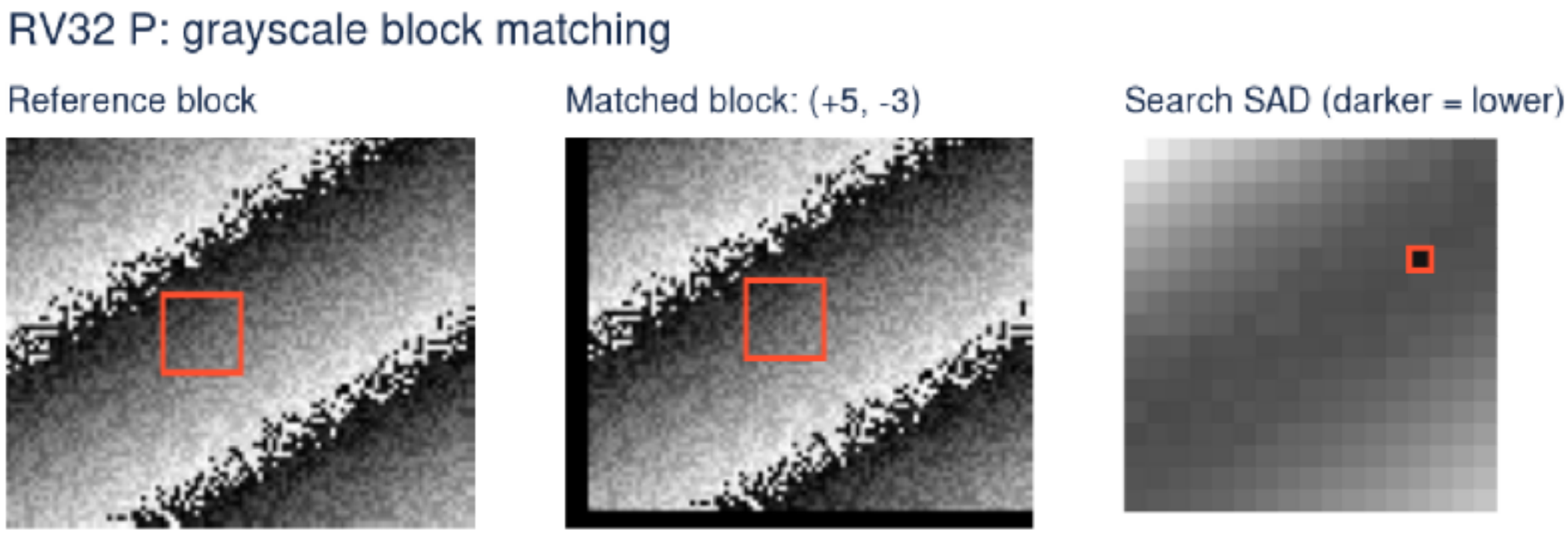
Image block matching, now on RV32 P

Add absolute pixel differences. The lowest SAD identifies the match.



289/289 image costs independently verified · best SAD = 0

Four pixels per packed operation

The image kernel is C. No inline assembly is required.

```
#include <riscv_packed_simd.h>

uint64_t sum = 0;
size_t x = 0;
for (; width - x >= 4; x += 4)
  {
    uint8x4_t av = __riscv_pld_u8x4 ((uint8_t *) (a + x));
    uint8x4_t bv = __riscv_pld_u8x4 ((uint8_t *) (b + x));
    sum += __riscv_pabdsumu_u8x4_u32 (av, bv);
  }
```

LOAD

Four pixels from each image

REDUCE

Four absolute byte differences

ACCUMULATE

Add to a 64-bit total

Generated packed SAD: pabdsumu.b / pabdsumau.b

Full kernel: row strides, alignment handling and scalar tails. RV64 uses eight byte lanes.

DSP support for a compact integer core

Packed arithmetic in the integer registers, using the same C source structure.

RV32 TARGET

RV32IMC + Zba/Zbb + P

P draft 0.6 · ILP32 soft-float ABI

Linux demo runtime: RV32IMAC.
RV64 counterpart: eight byte lanes.

HARDWARE LOAD PATH

Direct word loads for cores
with misaligned-access support.

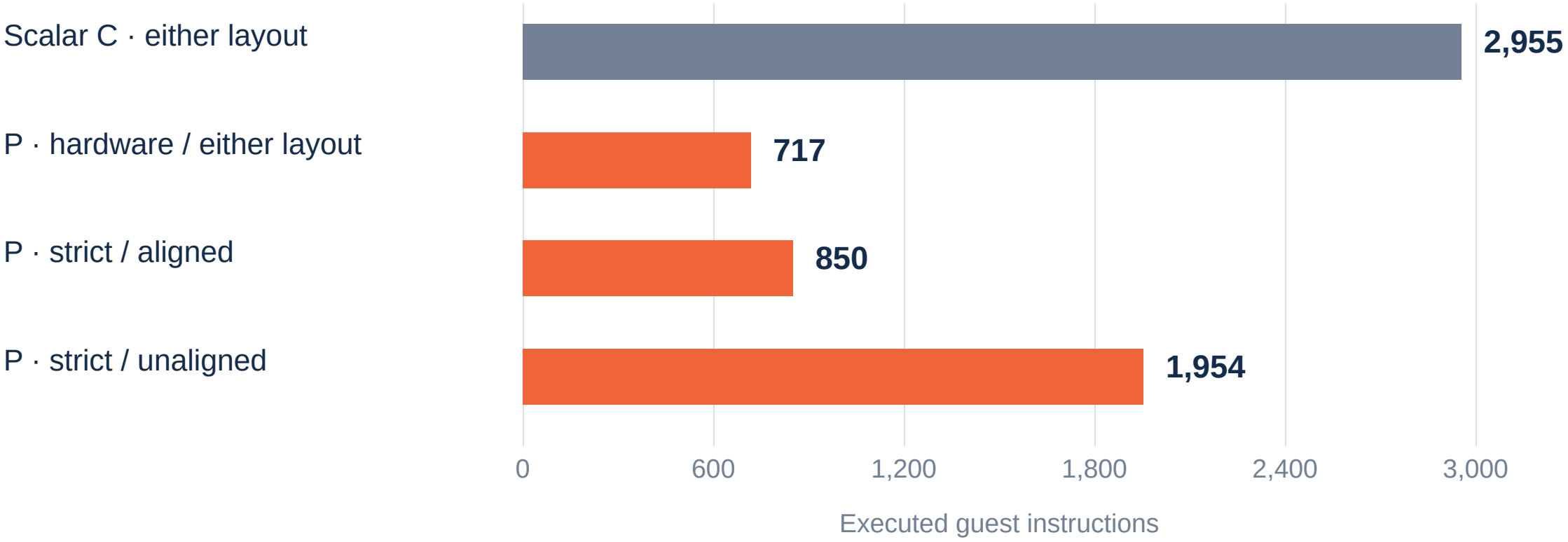
CHECKED LOAD PATH

Alignment checks select word loads
or byte loads plus packing.

Choose the packed width and load policy that fit your core.

RV32: 76% fewer executed instructions

One 16 × 16 kernel call, including return · -O3 · lower is better



Identical 16 × 16 pixels in both layouts; SAD = 22,364.

Choose code size or instruction count

RV32 hardware mode · kernel text and static frame sizes

Metric	Scalar · -O3	P · -O3	Scalar · -Os	P · -Os
Guest instructions	2,955	**717**	3,334	**845**
Kernel text, bytes	108	**204**	102	**132**
Static frame, bytes	0	**32**	16	**16**

P at -Os: text 204 → 132 bytes; instructions 717 → 845.

Strict P at -O3: 400 text bytes; 48-byte static frame.

A demo you can reproduce

C kernels, included tools, independently checked image-search costs.

12	82,640	289/289
configurations pass	C checks per build	image costs verified

RV32 / RV64 × hardware / strict alignment × O2 / O3 / Os

```
XLEN=32 ALIGNMENT=hardware bash scripts/present.sh
```

Try image matching, dot product and FIR with the included tools.

github.com/pz9115/RVP-Example

QEMU guest instructions, -O3/-Os, -mtune=size; hardware mode assumes misaligned word loads. Hardware timing not measured.